HUMAN COMPUTER AND INTERACTION (IT351)

ASSIGNMENT 1

Faculty: G. Ram Mohana Reddy

Semester: 6th sem B.Tech

Department of Information Technology,

National Institute of Technology Surathkal, Karnataka

Mangalore, India.

Submitted by:

Harsha K(15IT211)

NOV 2017

Microsoft Kinect

Kinect is a line of motion sensing input devices that was produced by Microsoft for Xbox 360 and Xbox One video game consoles and Microsoft Windows PCs. Based around a webcamstyle add-on peripheral, it enables users to control and interact with their console/computer without the need for a game controller, through a natural user interface using gestures and spoken commands **Features:**

- Improved body, hand and joint orientation

With the ability to track as many as six people and 25 skeletal joints per person—including new joints for hand tips, thumbs, and shoulder centre and improved understanding of the soft connective tissue and body positioning, you get more anatomically correct positions for crisp interactions, more accurate avateering, and avatars that are more lifelike.

- Microsoft Store support

You can now create Kinect enabled Windows apps by using familiar Windows Runtime components.

- Unity Pro support

For more than just gaming, Unity Pro offers cross-platform rapid prototyping.

- Powerful tooling

Kinect Studio provides enhanced recording and playback, and Visual Gesture Builder lets developers build their own custom gestures that the system recognizes and uses to write code by using machine learning, increasing productivity and cost efficiency.

- Advanced facial tracking

Resolution is 20 times greater, enabling the application to create a mesh of more than 1,000 points for a more accurate representation of a person's face.

Specifications:

Viewing angle: 43° vertical by 57° horizontal field of view

Vertical tilt range: ±27°

Frame rate: 30 frames per second

Audio format: 16-kHz, 24-bit mono pulse code modulation (PCM)

Audio input characteristics: A four-microphone array with 24-bit analog-to-digital converter (ADC) and Kinect-resident signal processing including acoustic echo cancellation and noise suppression

Accelerometer characteristic**s:** A 2G/4G/8G accelerometer configured for the 2G range, with a 1° accuracy upper limit.

Pros:

1. Easy setup and tailormade customisation
2. No controllers so no memorizing buttons
3. A lively gaming sense **Cons:**

1. Requires too much room
2. Expensive
3. Excessive hand-hovering
4. Voice recognition locale issues

Neyya (Fin Ring)

Fin is a small wearable of its kind, trendy gadget that you can wear on your thumb, which helps you to control your entire digital world. It uses smart Low Energy Technology such as Bluetooth for communication with connected devices such as Smartphones, TVs, automobiles and home automation devices.

Features:

- Three different device compatibility with single fin
- Recognizes each division of fingers
- Fin acts as security authenticator
- Gesture interaction on palm
- Fashionable
- Waterproof, dust proof, durable

Specifications: Waterproof

Size/fit: Flexible

Battery: Li Po custom battery with micro-USB charging dock; lasts a week;

Sensors: Fin can track accurate movement of your thumb on the palm also detect each part of your finger where you are touching

Communication: Bluetooth 2.1+ EDR and 4.0

Compatibility: Windows, MAC, iOS, Android, Windows Phones.

Pros:

- Tiny hence high portability
- Light, dustproof, waterproof and durable
- Affordable and acts as security authenticator
- Big asset for the visually challenged

Cons:

- Less amendment to specifications to be expected as it is a start up and has less funding
- Easy to replicate
- Easy to lose as it is tiny and no tracker in case of loss

Myo Bands

The Myo armband is a gesture recognition device worn on the forearm and manufactured by Thalmic Labs. The Myo enables the user to control technology wirelessly using various hand motions. It uses a set of electromyographic (EMG) sensors that sense electrical activity in the forearm muscles, combined with a gyroscope, accelerometer and magnetometer to recognize gestures. The Myo can be used to control video games, presentations, music and visual entertainment. It differs from the Leap Motion device as it is worn rather than a 3D array of cameras that sense motion in the environment. **Features:**

- a Bluetooth 4.0 low energy connection
- proprietary muscle activity electromyography (EMG) sensors
- an ARM processor
- 9-axis inertial measurement unit (IMU)
- a rechargeable lithium-ion battery
- Micro USB charging
- haptic feedback

Specifications:

- Arm size: Expandable between 7.5 - 13 inches (19 - 34 cm) forearm circumference
- Weight: 93 grams
- Thickness: 0.45 inches
- Sensors: Medical Grade Stainless Steel EMG sensors, Highly sensitive nine-axis IMU containing three-axis gyroscope, three-axis accelerometer, three-axis magnetometer
- LEDs: Dual Indicator LEDs
- Processor: ARM Cortex M4 Processor

- Haptic Feedback: Short, Medium, Long Vibrations
- Communication: Bluetooth

Pros:

- Simple setup
- Fun to use
- Substantial application library
- Ease of use

Cons:

- Unattractive design
- Sizing won't fit all
- Uncomfortable over long periods of time
- Gesture recognition needs improvement

Oculus Rift

The Oculus Rift is a virtual reality headset developed and manufactured by Oculus VR, a division of Facebook Inc., released on March 28, 2016.

Features:

- Field of Vision and Latency

  At its core, the Oculus Rift is a virtual reality (VR) headset, and this is not a new concept to the world of gaming technology. Its initial support will be for PC gaming, though future console support is being hinted at.

  The Rift features a 100 degree diagonal field of vision, which is much wider than is usually found on traditional VR headsets.

  The second feature is latency, the Rift is touted to support much lower latency than competing product, resulting in an experience that tracks head movements in a natural way.

- Game Support

  The team at Oculus Rift have been intelligent in being aggressive in building game support early on, particularly with the first-person-shooter genre of games that would be best served by a VR gaming product.

- Not Vaporware

  One of the most important features of the Oculus Rift is that it genuinely went to market. Many highly anticipated Kickstarter projects have featured attention-grabbing sales pitches, but floundered in implementation and going to market.

Specifications:

Product type: VR Headset

Sensors: Accelerometer, Gyroscope, Magnetometer, 360-degree positional tracking

Field of View: 110 degrees Resolution

per Eye: 1080 x 1200

Pros:

- Provides greater immersion into gameplay
- Can be utilized for healthcare applications as a therapeutic device ● Development kit cheaper to buy than most VR systems today.
- Rendering quality is good
- Latency with orientation detection is good

Cons:

- Prolonged use can cause eyestrain
- Without properly configuring the game first, you begin to feel nauseous the moment you start up the game
- The view for each eye lens can end up subtly different (shadow or reflection on one eye, but not the other) breaking immersion quite a bit.